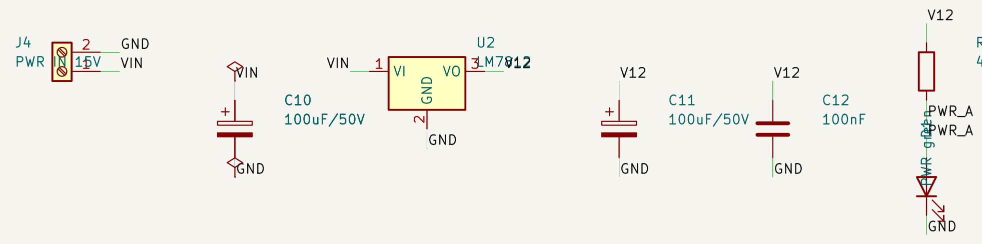
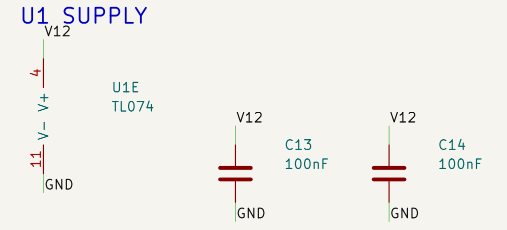
Bose Companion 5 sub crossover -- mono-summing 2nd-order Sallen-Key low-pass.

Single +12 V rail, no negative supply. All AC referenced to VGND = 6 V (A3 buffers the R8/R9 mid-rail divider).

A POWER IN -> +12 V

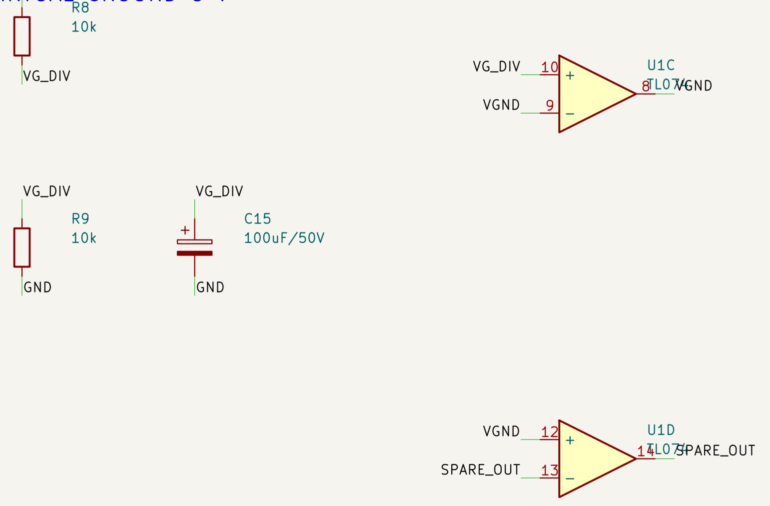


B U1 SUPPLY



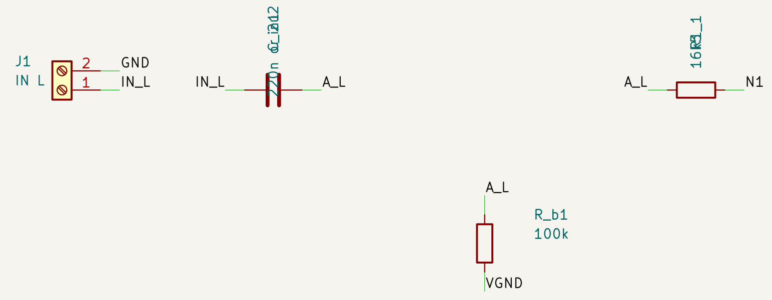
U1 supply: pin 4 = V12, pin 11 = GND. No negative rail.

C VIRTUAL GROUND 6 V



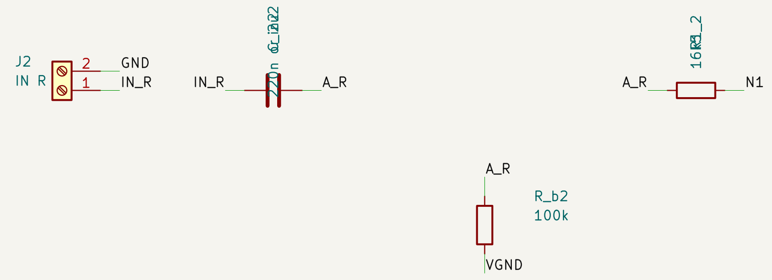
A4 is the spare section: + on VGND, out tied to in-.
An unterminated spare oscillates and couples into
its neighbours through the shared supply.

D INPUT L

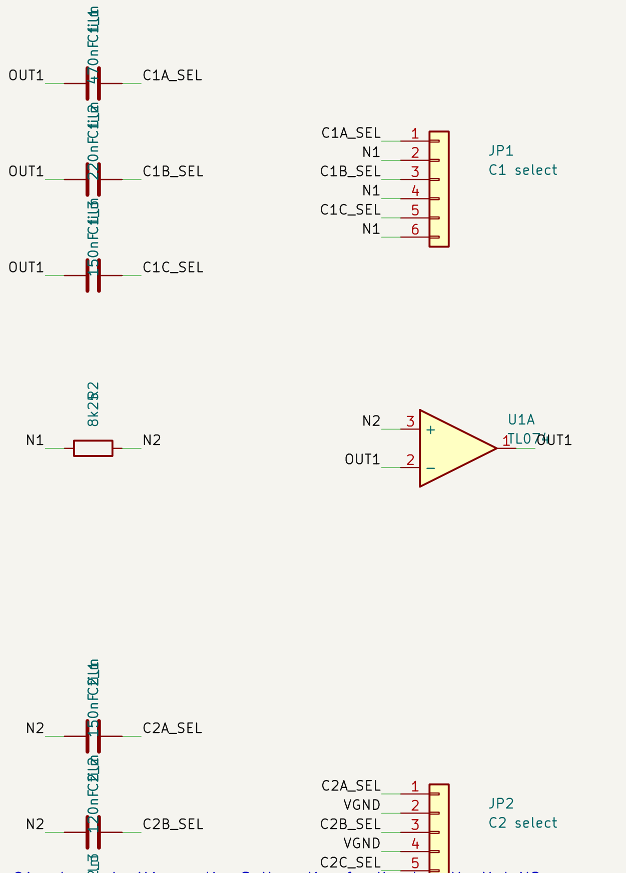


R_b1/R_b2 are the only DC path to A1's + input (C_in blocks DC,
C2 is a capacitor). Without them A1 drifts to a rail. They sit
after the coupling caps -- 100k at N1 would shift every corner by 8%.

E INPUT R



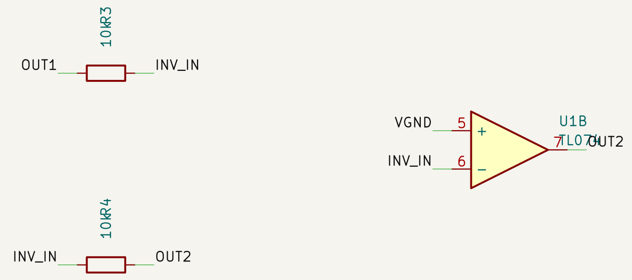
F SALLEN-KEY LOW-PASS



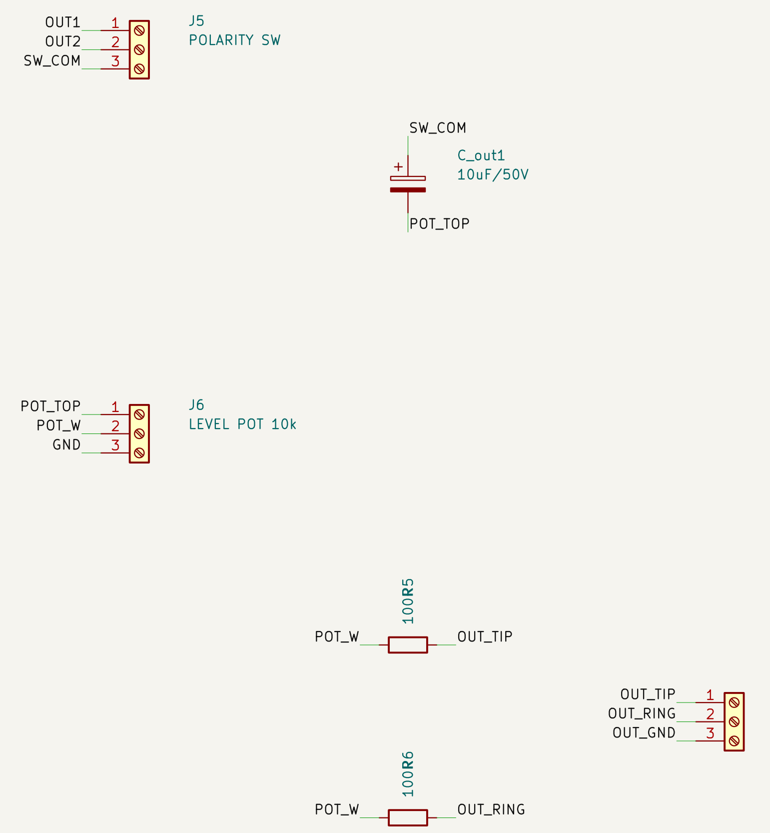
C1 returns to N1 -- the Sallen-Key feedback path. Not N2.
JP1 selects C1 (odd pin = cap, even pin = N1); JP2 selects C2 (odd = cap,
even = VGND). They select INDEPENDENTLY -- nine corners, not three pairs.
Corners below are MEASURED on the rev A board, not calculated: the quoted
f0 = 1/(2*pi*R*sqrt(C1*C2)) is not a -3 dB point, and C_in sits inside the
filter. See Results - Sub Crossover Bring-up.

Intended use is a panel 2-pole 3-position rotary, one pole per header:
pos 1 94 Hz A0->JP1 even, A1->JP1.1 (470n) B0->JP2 even, B1->JP2.1
pos 2 136 Hz A2->JP1.5 (150n) B2->JP2.3
pos 3 189 Hz A3->JP1.3 (220n) B3->JP2.5
NEVER wire one header pin to two switch lugs unless both lugs want the same
capacitance -- the lugs are then tied together and every position collapses
to the same value. This is why C1_1 is fitted rather than paralleling
C1_2 || C1_3 on one lug.
Shunts across a pin pair still work as a bench fallback.

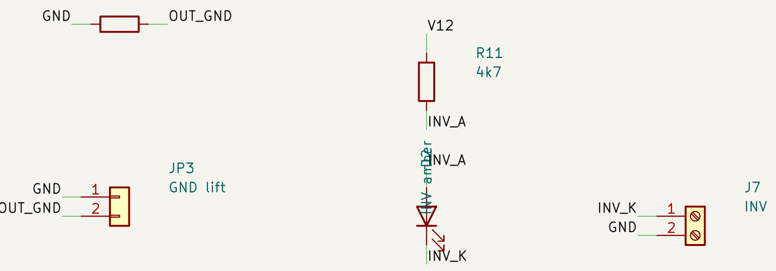
G POLARITY INVERTER



H OUTPUT STAGE



I STATUS LEDS



C_out + faces SW_COM: that node sits at 6 V DC, the pot side at 0 V.
J6 pin 3 = GND, not VGND -- C_out has already removed the 6 V bias;
referencing the pot to VGND would thump the driver at power-on.
JP3 shorts R7 (ground lift). Normally fitted.

D1 = power lamp on V12 (proves the LM7812, not just VIN).
D2 = INVERTED lamp, fed from SW2's unused second pole via J7.
Both ~2 mA. No LED touches a signal node.

C_in1/C_in2 use a dual-pitch footprint: 5.00 mm for the 220n as built (or a
2u2 non-polarised electrolytic), 15.00 mm for a real 2u2 film part. Fit ONE
pair. 220n puts ~8.8k of reactance in series with R1 at 82 Hz, so it is part
of the filter, not just an input high-pass -- it costs 1.3-3.1 dB of passband
level and damps the peaking. Whether that is a defect or a feature is a
listening question; Gate 11 has not answered it, so both pitches are on the
board and the envelope reserves room for the larger part either way.

Off-board: level pot (J6), polarity switch (J5), all panel jacks (J1/J2/J3).
Process: single-sided THT, bottom copper only, board outline 100 x 100 mm.
Copper is isolation-milled on the CNC with an 0.8 mm flat end mill; the fiber
laser does the silkscreen only. Routing clearance 0.85 mm / track 1.0 mm.
The 0.8 mm tool width is a hard floor on every copper-to-copper gap. Fixed-pitch
pads sit below the 0.85 mm target and carry a .kicad_dru exception: DIP-14
0.94 mm (passes), 2.54 mm headers and the TO-220 0.84 mm (do not).
Drills: 0.8 mm x52, 1.0 x26, 1.1 x3, 1.2 x4, 1.5 x15.

Reference names vs the handoff document: KiCad treats a reference with no
trailing digit as unannotated, so C1a/C1b/C1c -> C1_1/C1_2/C1_3,
C2a/C2b/C2c -> C2_1/C2_2/C2_3, R1a/R1b -> R1_1/R1_2, C_out -> C_out1.

Single-sided THT, CNC isolation milling (0.8 mm end mill), laser silkscreen
Mono-summing 2nd-order Sallen-Key low-pass, single +12 V rail

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Title: Bose Companion 5 sub crossover

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